Language Models are Few-Shot Learners

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Abstract

Recent work has demonstrated substantial gains on many NLP tasks and benchmarks by pre-training on a large corpus of text followed by fine-tuning on a specific task. While typically task-agnostic in architecture, this method still requires task-specific fine-tuning datasets of thousands or tens of thousands of examples. Here we show that scaling up language models greatly improves task-agnostic, few-shot performance, sometimes even reaching competitiveness with prior state-of-the-art fine-tuning approaches. We train GPT-3, an autoregressive language model with 175 billion parameters, 10x more than any previous non-sparse language model, and test its performance in the few-shot setting.

1. In-Context Learning

Our goal is to investigate whether scaling parameters alone enables in-context learning, where the model is conditioned on natural language descriptions of the task and a few prompt demonstrations without performing any parameter updates or weight modifications.

GPT-3 Architecture & Scaling

Section 2: Architecture Specifications across Parameter Scales

2.1 Model Specifications Table

Specifications for the eight distinct GPT-3 model sizes evaluated in the scaling study:

Model Name	Total Params	Layers	d_model	Heads	Batch Size (Tokens)
GPT-3 Small	125M	12	768	12	0.5M
GPT-3 Medium	350M	24	1024	16	0.5M
GPT-3 Large	760M	24	1536	16	0.5M
GPT-3 XL	1.3B	24	2048	16	1.0M
GPT-3 2.7B	2.7B	32	2560	32	1.0M
GPT-3 6.7B	6.7B	32	4096	32	2.0M
GPT-3 13B	13.0B	40	5120	40	2.0M
GPT-3 175B	175.0B	96	12288	96	3.2M

2.2 Training Compute Scaling Law

GPT-3 175B was trained on 300 billion tokens from Common Crawl, WebText2, Books1, Books2, and Wikipedia using 3.14 x 10^23 FLOPS of compute.

Performance & Limitations

Section 4: Empirical Findings and Critical Constraints

4.1 SuperGLUE Benchmark Performance

On the SuperGLUE benchmark, zero-shot GPT-3 175B achieves 71.8%, while few-shot (32 examples in context) achieves 75.6%, closing the gap with fine-tuned BERT-large models while requiring zero gradient steps on the target task.

5. Limitations & Societal Impacts

GPT-3 has noticeable limitations: it struggles with semantic comparison tasks (e.g. WiC), physical common sense reasoning, and long-range coherence beyond its 2048-token context window. Furthermore, sampling can produce factually incorrect hallucinations, and fine-tuning remains expensive due to model scale.